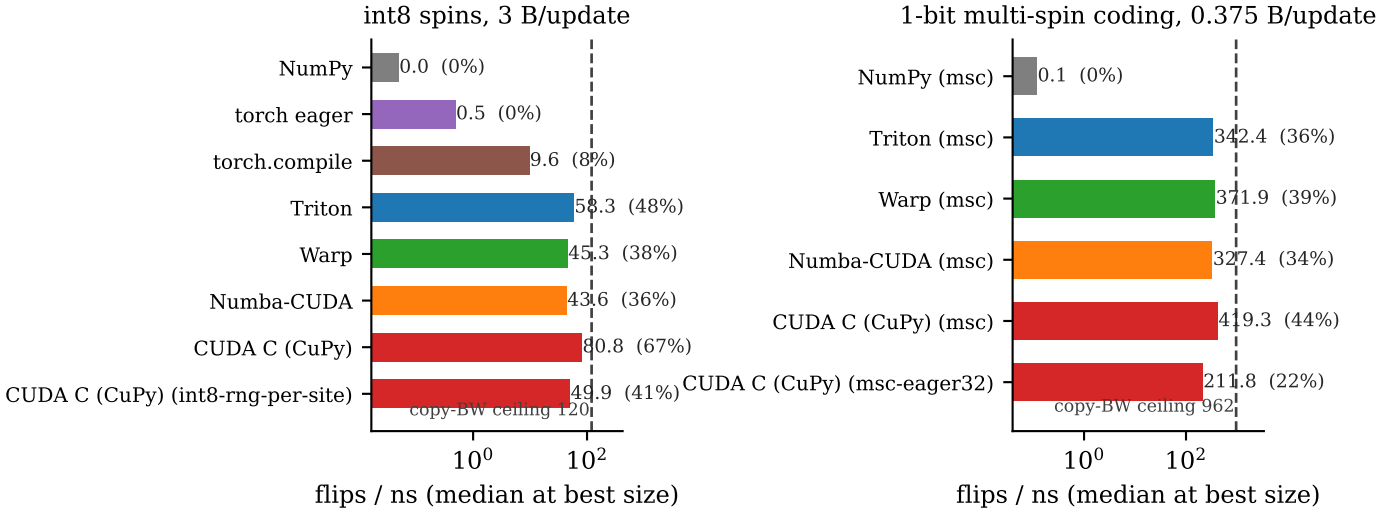
int8 spins, 3 B/update
NumPy
torch eager
torch.compile
Triton
Warp
Numba-CUDA
CUDA C (CuPy)
CUDA C (CuPy) (int8-rng-per-site)
0.0 (0%)
0.5 (0%)
9.6 (8%)
58.3 (48%)
45.3 (38%)
43.6 (36%)
80.8 (67%)
49.9 (41%)
copy-BW ceiling 120
10^0
10^2
flips / ns (median at best size)
1-bit multi-spin coding, 0.375 B/update
NumPy (msc)
Triton (msc)
Warp (msc)
Numba-CUDA (msc)
CUDA C (CuPy) (msc)
CUDA C (CuPy) (msc-eager32)
0.1 (0%)
342.4 (36%)
371.9 (39%)
327.4 (34%)
419.3 (44%)
211.8 (22%)
copy-BW ceiling 962
10^0
10^2
flips / ns (median at best size)